

Public Hand: Proximity-Aware Manipulation of Robotic Hand Shared with Multi-User in Robotic Room

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Figure 1: Overview of Public Hand: A robotic room is equipped with a cable-driven graspable hand hanged from ceiling, which provides a user with dynamic and flexible manipulation of the extended body part. Our proximity-aware approach is designed to achieve multiple user scenarios in public space with reduced conflict, which allows seamless hand-over of the maneuverability from a user to another.

Abstract

Robotic agents are increasingly appearing in our living spaces and are beginning to serve as partners to human users. While some robotic systems are expected to function autonomously, others are designed to operate under human control or as extensions of human users. This trend has led to a growing demand for effective manipulation methods for such robotic systems. Particularly in public spaces with multiple human users, manipulation methods must be designed to enable dynamic and flexible control while preventing conflicts among users.

This paper presents Public Hand, a proximity-aware robotic hand designed for shared use in a robotic room. Suspended from the ceiling and cable-driven, Public Hand extends the user's hand movements, enabling intuitive manipulation. By dynamically adjusting control based on social distance, it allows multiple users to interact naturally without wearable devices, facilitating smooth transitions between users. We introduce the system's design, control mechanism, and application scenarios through demonstration.

CCS Concepts

• **Human-centered computing** → **Interaction devices**; • **Computer systems organization** → *Robotic control*.

Keywords

Robotic Room, Cable-Driven Parallel Robotics, Shared Control, Proxemic Interaction, Human Robot Interaction

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1 Introduction

With the advancement of IoT and AI, interactions between various devices and humans within a space have been extensively explored. Traditional intelligent environments have mainly relied on information display devices such as monitors and speakers. However, in recent years, robots have also been incorporated as key components, enabling dynamic modifications of the space beyond mere information display.

To facilitate interaction, proxemic interaction, which leverages social distance (e.g., distance, orientation, and gaze) as an input modality, has emerged as a promising concept for spatial UIs that

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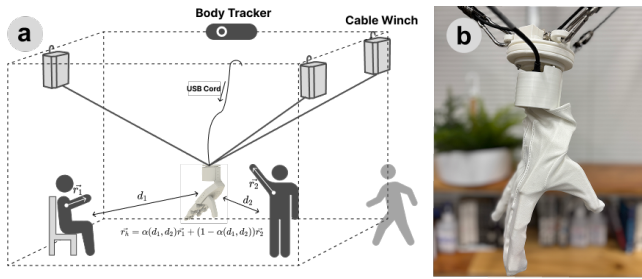


Figure 2: Public Hand System: (a) system architecture of cable-driven robotic hand (b) close-up of hand mechanics

adaptively respond to users in multi-user environments [Greenberg et al. 2011]. Building on this concept, there is growing demand for methods that enable seamless manipulation of spatially distributed robots in shared spaces.

Sato et al. proposed the concept of Robotic Room [Sato et al. 1996], a technology-integrated space composed of sensors and actuators. In such spaces, robot control has typically been command-based, using language or gestures as an input, treating the robot as a third party rather than an extension of the user and limiting flexible and dynamic manipulation. To address this, approaches like MetaLimbs [Sasaki et al. 2017], which use foot motion to reflect user intent in robot movement, have been explored. While it is promising for intuitive control, these methods face challenges in shared environments, where multiple users may conflict. Thus, an adaptive control mechanism enabling intuitive multi-user manipulation of a single robotic hand is needed.

To address this challenge, we apply the concept of proxemic interaction to Robotic Rooms. Specifically, we introduce a control mechanism where the influence over the robotic system dynamically changes based on the social distance between the users and the robot. To this end, we propose Public Hand, a graspable robotic hand that responds to human right-hand movements within the space (Fig. 1). The key features of Public Hand are as follows: (1) it is synchronized with the user's body, allowing intuitive operation as an extension of the right hand; (2) it requires no wearable devices, ensuring a non-intrusive interaction; and (3) it dynamically adjusts control based on the presence of multiple users in the shared environment. This approach enables people who may find wearable robots difficult to use (e.g., older adults) to operate the robotic hand while also allowing multiple users to engage in natural collaborative interactions within a shared space.

2 Public Hand System

Here we describe the system of Public Hand, designed to achieve motion control that reflects the intentions of multiple users in a shared space by leveraging social distance as an input.

When a user enters the designated area and slightly raises their right hand, the system recognizes that the user is connected to Public Hand. This mechanism ensures that only the intended user is allowed to operate the device.

Public Hand functions as an extension of the user's right hand by spatially extending the hand's movement at a fixed ratio relative to the user's position.



Figure 3: Example life scenarios with Public Hand: (a) holding baggage as an additional hand, (b) mediating the handover of flowers between two people (c) collaborative manipulation by two people to place a bouquet on an interior plant.

When multiple users are simultaneously connected, the system weights each input and reflects these in the overall motion. The weighting is determined based on the distance between each user and Public Hand; the input from the user closest to Public Hand has a stronger influence. Specifically, the gesture of the hand is governed by the movements of the user nearest to Public Hand (Fig. 2a).

The user's posture and position are captured using a Kinect sensor, and hand gestures are detected via MediaPipe in our first prototype. Public Hand is actuated by winch motors originally used for spatial lighting, controlled in real time by three ceiling-mounted lifts. To ensure safety, we designed a mechanical fuse that automatically disconnects the cable from Public Hand when excessive tension is applied (Fig. 2b).

3 Experiences

At SIGGRAPH 2025 Emerging Technologies Exhibition, we will demonstrate Public Hand equipped with the proposed proxemic manipulation system. Public Hand can serve as an extension of the hand for any user, even in environments populated by an unspecified number of individuals. Consequently, the system supports experiences not only in which a single user can utilize it as an additional hand (Fig. 3a), but also in scenarios that mediate handover interactions between two users (Fig. 3b) and facilitate collaborative operation by two users (Fig. 3c).

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